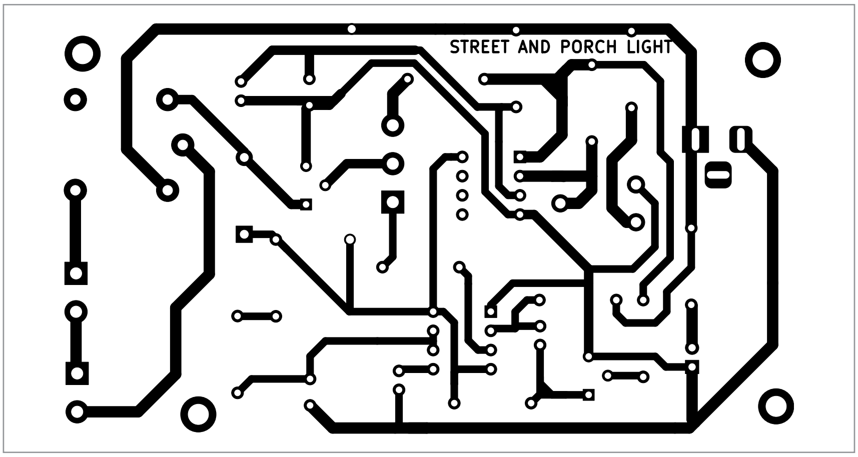


Fig. 3: Actual-size PCB of the street-cum-porch light

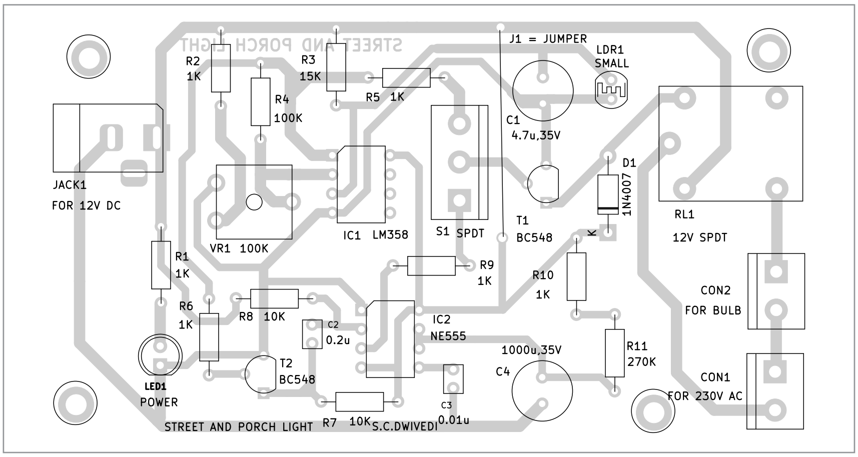


Fig. 4: Component layout of the PCB

deciding components are resistors R10 and R11 as ‘R’ and capacitor C4 (1000µF) as ‘C.’ These time-deciding RC can be set to around five minutes to switch on the porch light.

The transistors and relay RL1 are used to switch the street and porch lights on and off based on the output of the LM358 as well as the output of the NE555. A common relay RL1 is used to drive both the street light and the porch light.

## Construction and testing

An actual-size, single-side PCB layout of the street-cum-porch light circuit is shown in Fig. 3, and its component layout is shown in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box. The street-cum-porch light circuit will then be ready for use. After powering on the circuit, keep the SPDT switch S1 towards either street light or porch light based on the choice.

After assembling the components, install the unit near the porch, if it is to be used as a porch light. On the other hand, if it is to be used as a street light, install it near the street light.

***Bonus.*** You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/diy-video-street-light-cum-porch-light* EFY

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